

Algorithms Problem Solving

Level 3

Show All Clients



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Definition

File-Based Record Loading and Display

 Program reads all client records from a text file, converts each line into an `sClient`, and displays the records in a formatted table.

 It combines file reading, deserialization, and console formatting.

Explanation

File Reading

-  The client file is opened in input mode.
-  ``getline()`` reads the file one record line at a time.

Record Deserialization

-  Every valid line is passed to ``ConvertLineToRecord()``.
-  The serialized text is reconstructed into an ``sClient`` structure.

Tabular Formatting

-  Client fields are displayed using formatting tools such as ``setw()``.
-  Consistent column widths make the output easier to read.

End-of-File Processing

-  The preferred pattern is to process a line only when ``getline()`` succeeds.
-  This avoids the common problem of attempting to process an extra record after reaching the end of the file.

Procedures

1. Open the client file for reading.
2. Verify that the file opened successfully.
3. Print the table header.
4. Read each line using ``getline()``.
5. Convert each line into an ``sClient``.
6. Display the client's fields using formatted output.
7. Close the file.

💡 Example

Account Number | Pin Code | Client Name | Phone | Balance

➡ Result

A150	1234	Mohammed	079	5000.00
A151	5678	Ali	078	3200.00

Note

- ◆ ``setw()`` is useful for creating readable console tables.
- ◆ Empty or malformed lines should be handled before deserialization.
- ◆ File-reading logic can be separated from display logic for cleaner program design.

⊘ Warning

- ◆ Do not rely on `while (!MyFile.eof())` as the primary file-reading condition.
- ◆ Always check that the file opened successfully.
- ◆ Invalid records can cause parsing errors if field validation is not performed.



Comparison

▨ Loading and Displaying Clients

- ✓ Reads records from persistent storage.
- ✓ Deserializes each line.
- ✓ Presents the resulting structures to the user.

▤ Saving Clients

- ✓ Converts structures into serialized lines.
- ✓ Writes records to disk.
- ✓ Performs the opposite data flow.

✦✦ Summary

- ✓ Opens the client file for reading.
- ✓ Processes records one line at a time.
- ✓ Converts text records into structures.
- ✓ Displays all loaded clients in a formatted table.

Thank you 😊